

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Cryptography and Network Security

COURSE CODE: CSA5115

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 10%

QUESTIONS: 5

Total Marks:50

Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

I D. Gnana Deepthi(192411123) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: D. Gnana Deepthi

Annexure A

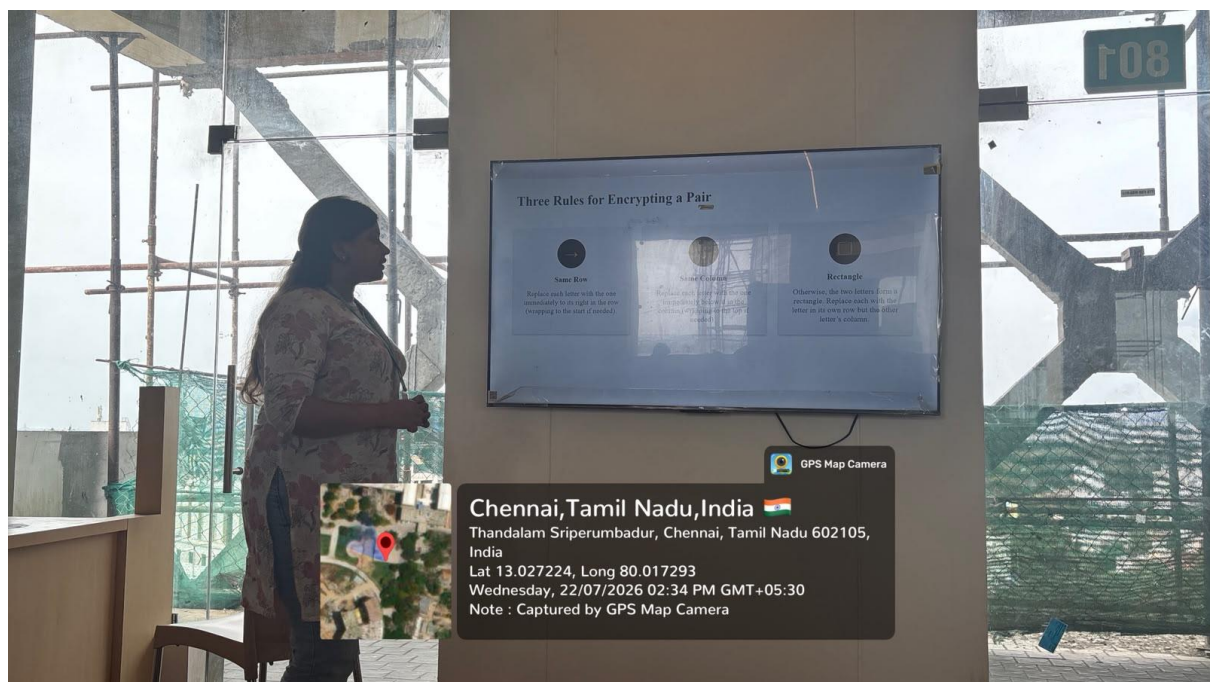
TECHNICAL PRESENTATION

1. From Caesar to Vigenère: Evolution of Substitution Ciphers”

2. “Substitution vs Transposition: Which Provides Better Security?” 3. “Security Attacks, Threats, Risks and Their Impact on Security Goals” 4. “Steganography vs Cryptography: Hidden Security in the Digital World” 5. “Role of Number Theory in Cryptography: From Euclid to Euler”

PLAYFAIR CIPHER PRESENTATION:

Good afternoon everyone. Today I am going to present on the Playfair Cipher, a classical encryption technique that encrypts pairs of letters instead of single letters, making it more secure than simple substitution ciphers. It uses a 5×5 key square created from a keyword such as MONARCHY. The plaintext is divided into pairs and encrypted using three rules based on the letters' positions in the key square. For example, the word HELLO is encrypted as CFSUPM. The same key square is used for decryption by applying the rules in reverse. Although the Playfair Cipher is no longer secure for modern communication, it is an important milestone in cryptography and helps us understand the basic concepts of encryption and decryption.



RUBRICS FOR CALCULATION:

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developin g	Level 1 – Beginning
Understan ding ng of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/tec hnologies	Good understanding ng with proper integration of most concepts	Basic understand ings g; limited or partial integration	Poor understand ing g; unable to integrat e concepts
Experime nt al Design	30	Well-planned, systematic implementation with optimal solution	Correct implemented ion with minor issues	Basic implement ation on with errors or inefficienci es	Incorrect or incomplet e implement ation on
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsisten t use of tools	Incorrect or minimal use of tools
Analysis of Result s	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interp retation n	Basic analysis with limite d insights	Poor or incorrect analysis of results

Documentation	10	Clear, well structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation
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